

Tunnel Survivor

Module	SE4031 - Games Development
Assignment	Assignment 02 (Individual) - 2D Interactive Game
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1. Game Concept

Sky Survivor is an endless-style 2D survival game where the player controls a flying character and tries to stay alive for as long as possible. The player moves vertically to pass through the gaps in tunnel obstacles while also dealing with enemies and collectible health packs.

Tunnel Survivor is a 2D endless survival game where the player needs to navigate through moving tunnel obstacles while avoiding enemies and picking up health packs. The goal is to survive for as long as possible and earn the highest score. The game has a side-scrolling feature where obstacles are always moving towards the player, making the game harder over time.

2. Core Feature Implementation

Player Movement

- Player can move vertically (up and down)
- Smooth movement using keyboard input
- Restricted within screen boundaries

Health System

- Player starts with a fixed health value
- Health decreases when:
 - Colliding with tunnel walls
 - Colliding with enemies
- Health increases when:
 - Collecting health packs
- Game ends when health reaches 0

Score System

- Score increases when player successfully passes obstacles
- Score is displayed in real-time on screen
- Final score is shown on Game Over screen

Enemies

- Enemies move toward the player
- Collision with enemies reduces player health
- Adds challenge and interaction

Health Packs

- Spawn randomly
- Restore player health when collected
- Encourages risk/reward gameplay

Shooting System

- Player can shoot projectiles
- Projectiles destroy enemies
- Adds action element to gameplay

Endless Loop System

- Obstacles, enemies, and health packs continuously spawn
- Game never stops until player dies
- Increasing difficulty over time

3. Controls

Input	Action
W / Up Arrow	Move player upward
S / Down Arrow	Move player downward
Space	Fire projectile
R	Restart after game over

4. Creative Feature

- Popup-style Game Over panel
- Color-changing health indicator (Green → Yellow → Red)
- Continuous gameplay (endless runner style)
- Player shooting system for interaction
- Enemy + obstacle combination for increased difficulty

5. Screenshots



Figure: Gameplay view showing the player, enemy, tunnel obstacles, and projectile.

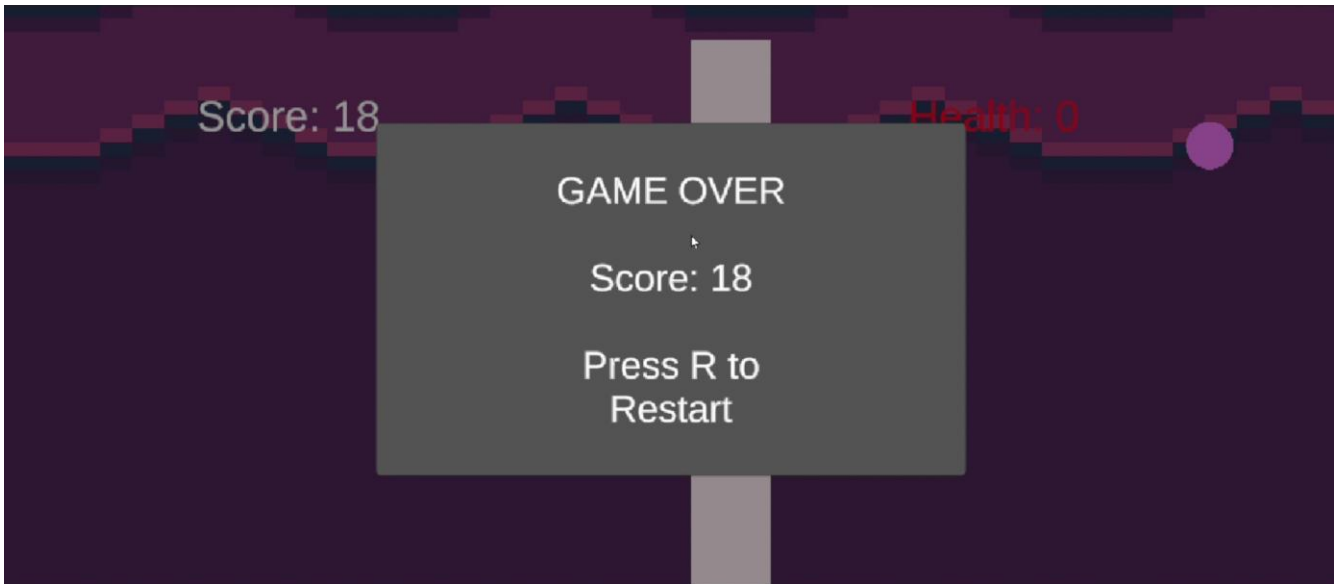


Figure: Game Over popup showing final score and restart instruction.

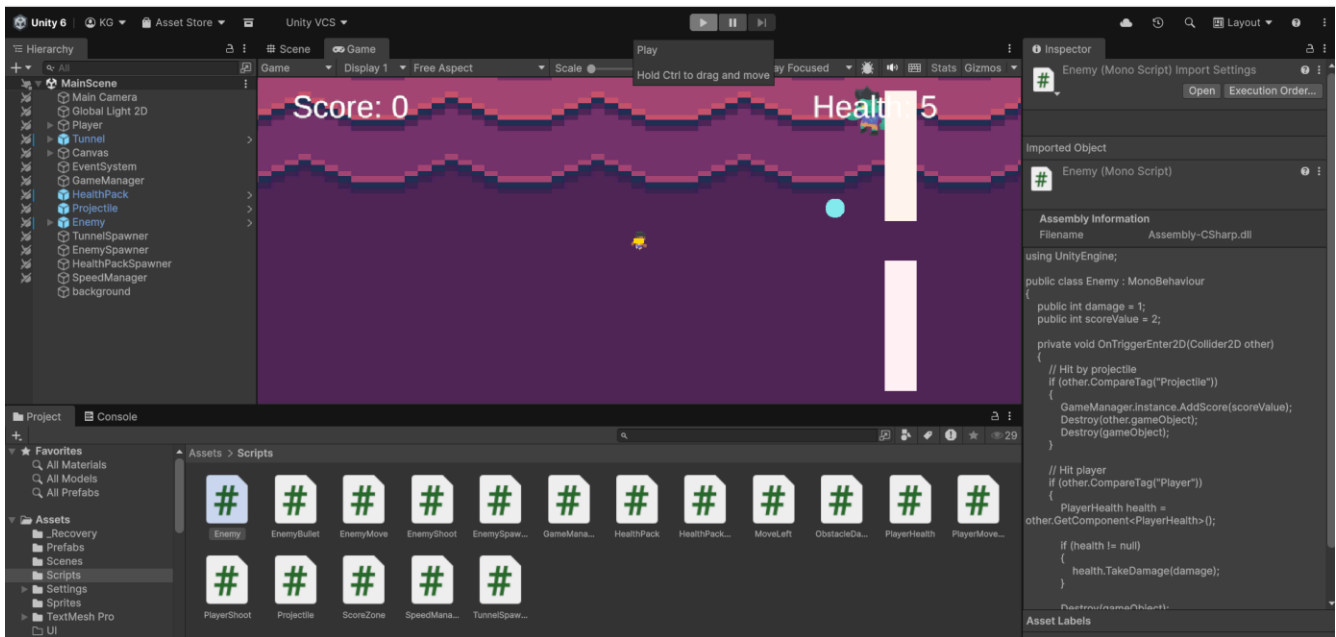


Figure: Scene setup used to position tunnels, and player lane.

6. External Assets and Tools

- Unity 6 was used as the main development environment.
- Kenney Toon Characters pack was used for character and enemy sprite enhancement.
- TextMeshPro was used for on-screen UI text.
- The final Windows build was prepared according to the assignment submission instructions.
- C# scripting
- Unity Physics (Collider & Trigger system)

7. Video Link & project zip file

<https://drive.google.com/drive/folders/1hi2K764JraiTw2LZKlI27mptEzpeyc8X?usp=sharing>